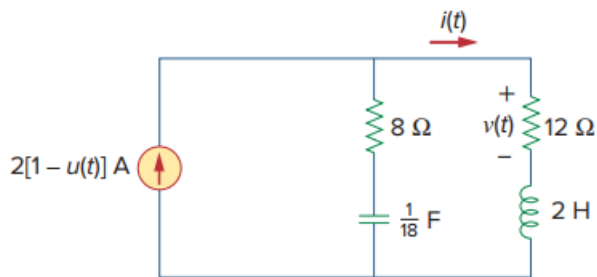


Problem

Given the circuit shown in Fig. 8.103, determine the characteristic equation of the circuit and the values for $i(t)$ and $v(t)$ for all $t > 0$.

Figure 8.103:



Step-by-step solution

Step 1 of 15

Refer to Figure 8.103 in the text book.

At $t < 0$, the capacitor acts as open circuit and inductor acts as short circuit. The value of the current source is 2 A since the value $u(t)$ at $t < 0$ is 0 .

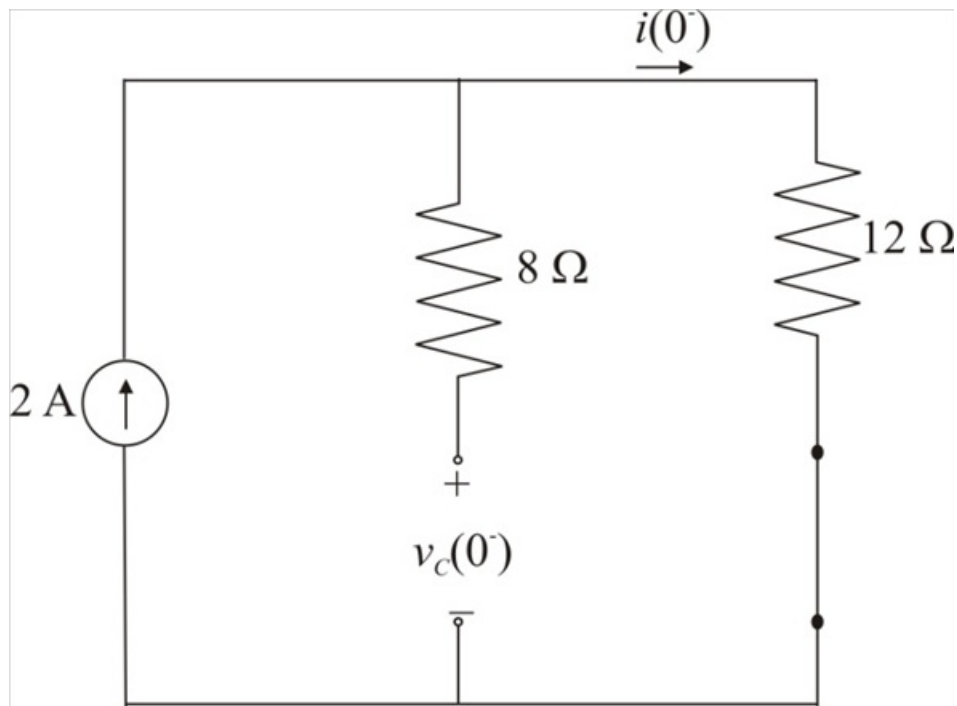


Figure 1

[Comments \(1\)](#)



Anonymous

This is totally wrong. Inductors act as open circuits and capacitors act as short circuits

Notice that no current passes through $8\ \Omega$ resistor.

The value of the current $i(0^-)$ is 2 A .

Calculate the value of the voltage $v(0^-)$.

$$v(0^-) = 12i(0^-)$$

Substitute 2 A for $i(0^-)$.

$$\begin{aligned} v(0^-) &= 12(2) \\ &= 24\text{ V} \end{aligned}$$

Therefore, the value of the voltage $v(0^-)$ is 24 V .

Step 3 of 15

The value of the current source at $t > 0$ is 0 A .

Draw the circuit diagram at $t > 0$.

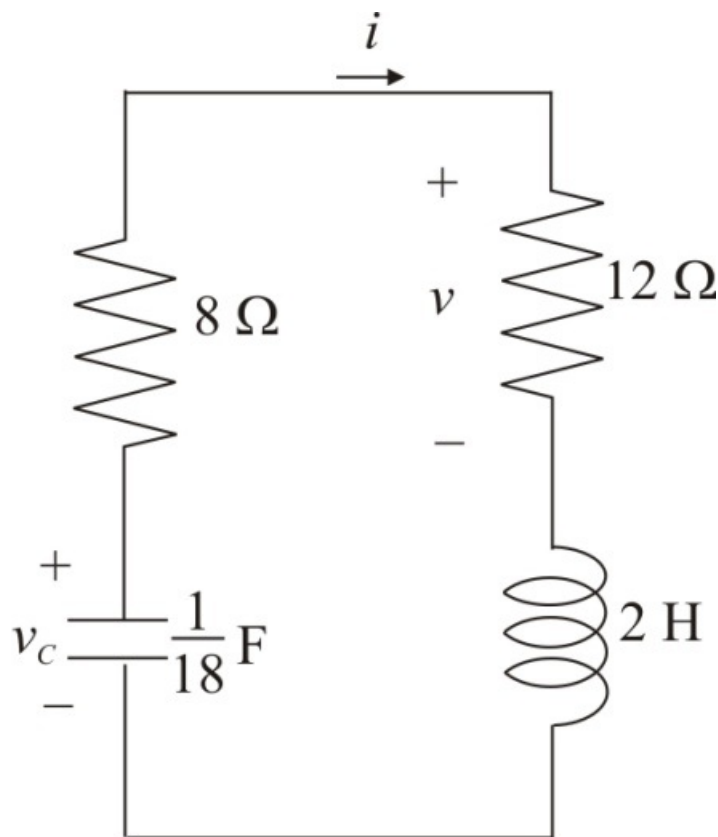


Figure 2

Step 4 of 15

The voltage across the capacitor and the current through the inductor will not change instantaneously. Hence, at $t > 0$ the value of the voltage across the capacitor $v(0^+)$ is 24 V and the value of the current $i(0^+)$ is 2 A .

Notice that the voltage across the capacitor and the current through the inductor are continuous. Therefore, the values at $t = 0$ are $v(0) = 24\text{ V}$ and $i(0) = 2\text{ A}$.

Step 5 of 15

Apply Kirchhoff's Voltage Law in loop.

$$12i + 2 \frac{di}{dt} - v_C + 8i = 0$$

$$2 \frac{di}{dt} - v_C + 20i = 0$$

$$2 \frac{di}{dt} = v_C - 20i$$

$$\frac{di}{dt} = \frac{v_C - 20i}{2}$$

Calculate the value of $\frac{di}{dt}$ for $t = 0$.

$$\frac{di(0)}{dt} = \frac{v_C(0) - 20i(0)}{2}$$

Substitute 24 V for $v_C(0)$ and 2 A for $i(0)$.

$$\begin{aligned} \frac{di(0)}{dt} &= \frac{24 - 20(2)}{2} \\ &= \frac{24 - 40}{2} \\ &= -\frac{16}{2} \\ &= -8 \text{ A/s} \end{aligned}$$

Therefore, the value of $\frac{di(0)}{dt}$ is -8 A/s .

Step 6 of 15

Apply Kirchhoff's Voltage Law in loop.

$$12i + 2 \frac{di}{dt} + \frac{1}{(1/18)} \int i \, dt + 8i = 0$$

$$20i + 2 \frac{di}{dt} + 18 \int i \, dt = 0$$

Apply derivative with respect to t .

$$2 \frac{d^2i}{dt^2} + 20 \frac{di}{dt} + 18i = 0$$

$$\frac{d^2i}{dt^2} + 10 \frac{di}{dt} + 9i = 0$$

Treat $\frac{d}{dt}$ as s .

$$s^2i + 10si + 9i = 0$$

$$(s^2 + 10s + 9)i = 0$$

Step 7 of 15

Consider the characteristic equation of the circuit.

$$s^2 + 10s + 9 = 0$$

$$(s + 9)(s + 1) = 0$$

Therefore, the roots of the equation are -9 and -1 .

The roots are negative and real, hence the circuit exhibits over damped response.

Step 8 of 15

Draw the circuit diagram for $t = \infty$.

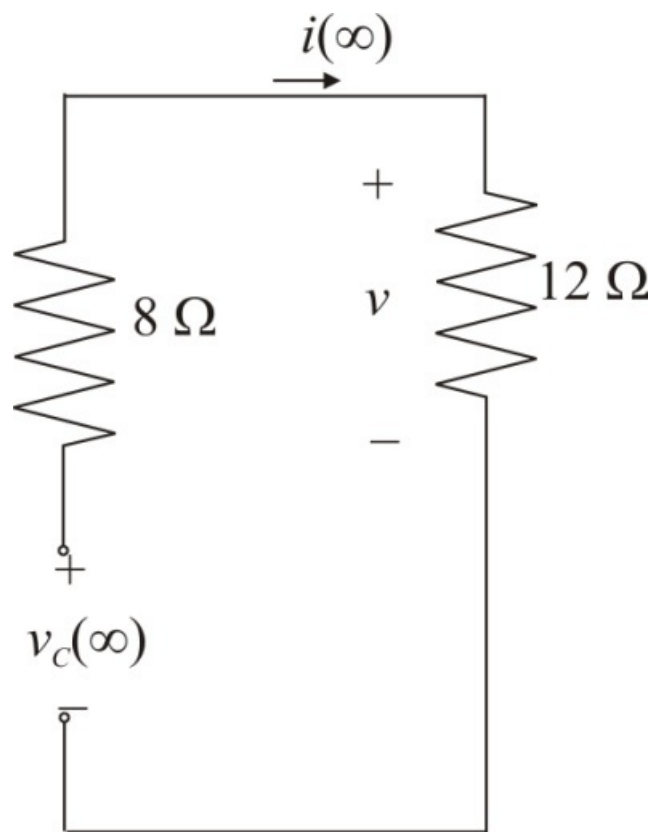


Figure 3

Step 9 of 15

Therefore, the value of the current $i(\infty)$ is 0 A .

Step 10 of 15

Consider the expression of the current $i(t)$.

$$i(t) = i(\infty) + A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Substitute 0 A for $i(\infty)$, -9 for s_1 and -1 for s_2 .

$$\begin{aligned} i(t) &= 0 + A_1 e^{-9t} + A_2 e^{-1t} \dots\dots (1) \\ &= A_1 e^{-9t} + A_2 e^{-1t} \end{aligned}$$

Step 11 of 15

Recall equation (1).

$$i(t) = A_1 e^{-9t} + A_2 e^{-1t}$$

Substitute 0 for t .

$$\begin{aligned} i(0) &= A_1 e^{-9(0)} + A_2 e^{-1(0)} \\ &= A_1 e^0 + A_2 e^0 \\ &= A_1 (1) + A_2 (1) \\ &= A_1 + A_2 \end{aligned}$$

Substitute 2 A for $i(0)$.

$$2 = A_1 + A_2$$

$$A_2 = 2 - A_1 \dots\dots (2)$$

Step 12 of 15

Recall equation (1).

$$i(t) = A_1 e^{-9t} + A_2 e^{-1t}$$

Apply derivative with respect to t .

$$\begin{aligned} \frac{di(t)}{dt} &= A_1 \frac{d(e^{-9t})}{dt} + A_2 \frac{d(e^{-1t})}{dt} \\ &= -9A_1 e^{-9t} - 1A_2 e^{-1t} \end{aligned}$$

Substitute 0 for t and $2 - A_1$ for A_2 .

$$\begin{aligned} \frac{di(0)}{dt} &= -9A_1 e^{-9(0)} - 1(2 - A_1) e^{-1(0)} \\ &= -9A_1 e^0 - 1(2 - A_1) e^0 \\ &= -9A_1 - 2 + A_1 \end{aligned}$$

Substitute -8 A/s for $\frac{di(0)}{dt}$

$$-8 + 2 = -8A_1$$

$$-6 = -8A_1$$

$$A_1 = \frac{6}{8}$$

$$A_1 = 0.75$$

Therefore, the value of the constant A_1 is 0.75.

Step 13 of 15

Recall equation (2).

$$A_2 = 2 - A_1$$

Substitute 0.75 for A_1 .

$$\begin{aligned} A_2 &= 2 - 0.75 \\ &= 1.25 \end{aligned}$$

Therefore, the value of the constant A_2 is 1.25.

Step 14 of 15

Recall equation (1).

$$i(t) = A_1 e^{-9t} + A_2 e^{-t}$$

Substitute 0.75 for A_1 and 1.25 for A_2 .

$$i(t) = 0.75e^{-9t} + 1.25e^{-t} \text{ A}$$

Therefore, the value of the current $i(t)$ is $0.75e^{-9t} + 1.25e^{-t} \text{ A}$.

Step 15 of 15

Calculate the value of the voltage $v(t)$.

$$v(t) = 12i(t)$$

Substitute $0.75e^{-9t} + 1.25e^{-t} \text{ A}$ for $i(t)$.

$$\begin{aligned} v(t) &= 12(0.75e^{-9t} + 1.25e^{-t}) \\ &= 9e^{-9t} + 15e^{-t} \text{ V} \end{aligned}$$

Therefore, the value of the voltage $v(t)$ is $9e^{-9t} + 15e^{-t} \text{ V}$.